COC to Host Class A Meet

Way back in April, six COC members went to a class A meet in Reno. It was an enjoyable time and many new friendships were made. There were a few problems, however. A marker common to the red and blue courses was hung incorrectly on the first day and a poorly mapped area was used on blue on the second day. This was quite unfortunate as some individuals, like the Larssons, would have been nationally ranked this year if these days had not been voided. Writing in the May-June Bay Area Orienteering Club newsletter, Joe Scarborough, BAOC President, observes "There is a lesson to be learned here. The West Coast clubs have not yet developed the manpower and expertise to put on a meet without major errors.... The result is some very disappointed orienteers who have travelled a long way.... Why not wait until we have enough experienced people in the West willing to put in the time required to stage a successful class A meet?"

Well, COC accepts the challenge. Work has begun on a class A meet to tenatively be held next October 3-4 near Cle Elum. Class A meets differ from the class B meets COC has hosted this year in two significant ways: 1) they carry national ranking points and 2) they have fairly strict requirements which must be met before the US Orienteering Federation will sanction them as class A (this last consideration is the reason for rescheduling the meet from April to next October - a lot of fieldwork is required to produce a top quality map).

There will be plenty of oppurtunity for volunter work before and during the meet. Help is needed with publicity, registration, timing, fundraising, mapmaking, and other tasks. Some, like timing, will not be needed until the meet itself while others (like mapmaking and fundraising) will be needed this winter or next spring. To gain more experience, four class B meets are planned between now and next fall. Everyone is encouraged to help run one or two of them - Cascade is a strong club and getting better.

—M. Schuh

Compass Mementos

Why should I
A normally sane and sensible guy,
Suddenly fly
Into the trees
And crawl for miles on hands and knees!
and Quo Vadis
I'm not sure whether I'm past it,
or whether I still haven't come to it yet.

~unknown

Time Dated Material

Cascade Orienteering Club
P.O. BOX 3572
BELLEVUE, WASHINGTON 98009

BEARING 315

Newsletter of the Cascade Orienteering Club

NOVEMBER/DECEMBER 1980

The BC Championship at Brandywine Falls, near Vancouver, was well attended by members of COC to the tune of about seventeen. The meet was a two-day event on September 6 and 7 in a runable open forest which as most of us know is not too common on the west coast. This photo is most of the COC members who were there, are you in the picture?

COC Members at US Championships

Open forest with little underbrush, rolling hills and good runalbe surfaces with few rocks characterized the terrain at the 1980 national championship orienteering meet. About 450 participants from all over the U.S. and several foreign countries attended. The autumn colors, cool but sunny weather, well organized events and entusiastic orienteers combined for a wonderful weekend.

Goran and Berith Larsson with their son Robert, Bill Hall and Dave Enger all made the trip from Seattle to the St Louis area. Göran, Bill and Dave ran the blue course along with about 51 other men. Göran placed 5th and Dave placed 28th, while Bill had some difficulty with the challenging courses, especially the first day. Berith placed very well on the red course.

Dieter Wolf, a world class orienteer from Switzerland, easily won the mens (H21A-Blue) competition. Eric Weyman of the Ramapo (New Jersey) O Club placed second and was declared the new U.S. man's champion. Peter Gagarin, the former and long time champion was 3rd. West coast competitors made a better showing this year with Dennis Holleran of Oregon in 4th, Göran in 5th and Bruce Wolfe of the San Francisco Bay Area O Club in 7th place. Sharon Crawford of the New England O Club easily won the D19A- red course competition to retain her long standing status as U.S. national women's champion.

Many of you remember Mike and Corrie Fiedler, last year's Cascade O Club president and secretary who now lives in Cedar Rapids, Iowa. They both ran the orange course, with Mike winning the H21C competition. Mike and Corrie belong to the Minnesota O Club, although because of busy schedules and long distances they are able to attend only a few events. We all enjoyed seeing each other again, but the weekend ended too soon.

—D. Enger

from the President

I'd like to make a pitch for the United States Orienteering Federation (USOF). The stated objectives of USOF
tive of USOF is "to promote and support Orienteering in the United States of America and internationally". USOF serves as the national clearinghouse for orienteering information and is a liaison with the International Orienteering Federation (IOF). USOF charters and assists local clubs, provides materials and expertise as well as coordinates and sanctions national events such as the Reno class A meet in April and last month's national championship meet.

The Cascade Orienteering Club is USOF chartered and several of us have individual or family USOF memberships. Almost anyone may support the sport by joining USOF. Benefits of membership includes receiving the monthly magazine Orienteering USA which contains many helpful mapping and competition tips as well as information on the national scene. For more information on USOF membership me, Mike Schuh or Clayton Smith (474-5675, Tacoma).

Dave

See page 7 for info on USOF

Cascade Orienteering Club

BEARING 315

PRESIDENT	Dave Enger	455-5320, 789-4137
VICE-PRESIDENT	Charlie Bock	641-4791
SECRETARY	Lorrie O'Brien	746-4[illegible]
TREASURER	Jean Hendrickson	772-1[illegible]
EVENTS	Len Englund	746-3822
PUBLICITY	Tony Paulson	784-4649
MEMBERSHIP	Mike Schuh	783-3960
MAPPING	Göran Larsson	938-2058
TRAINING	Gary & Debbie MacDonald	823-9337

NEW MAPS

Several mapping projects are underway or will commence soon. Map-making is a very good way to improve your map reading skills and your help is invited. If you would like to lend your talents on any of the following maps, just give the person in charge a call.

Carkeek Park	Dave Enger	789-4137
Discovery Park	Mike Schuh	783-3960
Lincoln Park	Debbie Newell	824-7250
	or Richard Lacy	937-1533
Fort Steilacoom	Clayton Smith	474-5675
Farrel McWhirter	Goeran Larsson	938-5675
Cle Elum Ridge	Goeran Larsson	

COC is always looking for new areas for orienteering, especially large ones that may be used year round. If you know of any likely **locations** - anywhere in the state with about 300 acres or more - please bring them to the attention of Goeran (Clayton in Tacoma) or any other officer or meet director.

Jean Davis is one happy orienteer as she punches a control at the Brandywine Falls, B.C. meet, site of the BC Championships

Richard Lacy pauses for the camera at a control located "at the foot of the cliff" in Brandywine Falls, B.C. (Richard that must have cost you at least 3 seconds!)

Editorial Comment

With the help of some articles from Dave and Mike this issue has grown by one page. Some of the meet results and those articles by Mike were done on his word processor and were sent to me photo ready - this newsletter is going places, why not be a part of it. Thanks Mike and Dave, I welcome any material for the newsletter. How do you like the photos? If you have any Black and White photos of any of our events consider sending them in to the newsletter and share them with us - I will insure that any photos sent in for printing will be returned the same way you sent them. I am looking forward to the next issue which will contain an article about our planned class A meet next year near Cle Elum.

Yours in orienteering

Clayton

Membership

If you would like to join in the fun and learn more about the sport of orienteering, we would be happy to have you as a member of the Cascade Orienteering Club. Fill in the membership blank below and send with the yearly dues to:

Cascade Orienteering Club
P.O. Box 3572
Bellevue, WA 98009

☐ **Renewal** ☐ **New Member**

☐ Family $8.00 ☐ Individual $5.00 ☐ Associate (under 18 years) $3.00

Name: _____

Address: _____

_____

Telephone Number: _____

As your friendly neighborhood Membership Coordinator, I would like to ask you to briefly check your mailing label. It includes your membership class and expiration date. If you find any errors - and there are probably a few - please bring them to my attention. Also, I would appreciate an assistant to help update our membership files about once every two months. If you are interested, have even the most meager of typing abilities, and live in the north end, please give me a call.

This will be the last newsletter of 1980 and I wish to remind some of you that your memberships will expire at the end of this year. If you wish to renew (or join), please send your fees to the club address by January 15. Gift memberships are also accepted - just let me know who the lucky person is.

The past 12 months have been a period of strong growth for our club. Considering only pure numbers, membership has nearly doubled to 124 (3 Life, 45 Family, 71 Single, and 5 Associate members) while our total mailing has more than tripled to over 300. It is this large membership base which allows us to hold two meets per month and produce color maps. 1981 should be just as good.

- Mike Schuh

United States Orienteering Federation

P O BOX 1039
BALLWIN, MISSOURI 63011

Materials Available from USOF Headquarters - 07/01/79

		Quan
USOF Bumber Sticker	$ .50	
USOF Patch	1.00	
USOF T-shirt (Small, Medium, Large, X-Large)	4.50	
USOF's Bylaws and Rules for National Competition	2.00	
IOF Map Drawing Specifications	2.30	
IOF Rules for International O Meets	1.40	
IOF Principles of Course Planning	.70	
Map Making For Orienteers (Harvey)-USOF members	not-	
-others	available	
Organizing Informal O Meets	.25	
Past Issues of Orienteering USA	.25	
Past Issues of Control Point	.15	
Trim Orienteering Pamplet	1.25	
Basic Orienteering Lesson Plans	.15	
O' Equipment Sources / Short Bibliography	.05	
Bibliography of Orienteering Publications	.10	
List of USOF-affiliated clubs	.05	
How to Organize an O Club	.10	
How to get News Coverage for Your O Meet	.10	
USOF membership application and club application	----	

Please send me the items checked in the quantities indicated. Attached is a check for those items ordered.

Name _____

Address _____

_____

_____ Zip _____

Note: Please add 25¢ or a couple of 15¢ stamps per order to cover postage and handling.

Issues of Control Point available: Jan-Feb 1978, June 1978, July-August 1978, October 1978, November 1978. Issues of Orienteering USA available: Spring 1974, Summer 1974, Summer 1975, Summer 1978. All issues of the merged OUSA/CP are available (1979).

United States Orienteering Federation

P. O. BOX 1039
BALLWIN, MISSOURI 63011

Application for Membership - Date _____

Please check - New Member ( ) Renewal ( )

I hereby apply for membership in the U.S. Orienteering Federation. I understand that I am entitled to all benefits of the organization, including membership in the International Orienteering Federation. I am applying as follows:

A. As a member of a club chartered with USOF:
Club _____
Club officer's signature _____
( ) Student $4 ( ) Individual $8
( ) Family $12 ( ) Lifetime $125
( ) Contributing $40 or more

B. Member-at-large (not a club member):
( ) Student $5 ( ) Individual $10
( ) Family $15 ( ) Lifetime $125
( ) Contributing $40 or more

Name (print) _____ Sex ____

Address _____

_____ Zip _____

Year of Birth _____ Home phone _____

(If family membership, please list other members)

Name _____ Sex ___ Yr born ____

_____ ___ ____

_____ ___ ____

Membership expires one year from date of receipt by USOF. For renewals, membership is for one year from the expiration date shown on the front of your card. Please make checks payable to the United States Orienteering Federation and mail to USOF at P.O. Box 1039, Ballwin, MO 63011.

Schedule of upcoming events

GENERAL INFORMATION - Unless otherwise noted, novice (white), intermediate (yellow or orange) and advanced (red or blue) courses will be available at all Cascade Orienteering Club meets. Registration and start times occur from 10:00am until noon and the fee is usually $1.00 per map for COC members and $2.00 for non members. Instruction for beginners is available at all events.

SATURDAY, NOVEMBER 22 - WOODLAND PARK NIGHT MEET and POTLUCK DINNER This will be our first experience with this interesting variation of orienteering. The terrain is quite open and the moon will be full. The control markers will be reflective and everyone must bring a good flashlight or headlamp. Registration will be 4:30 - 6:00pm in the picnic area inside the N 50th St. entrance, located just east of Aurora Ave. N in Seattle. Meet Director; Richard Lacy.

All participants are welcome to a post meet potluck dinner, beginning about 7:00pm at Dave Enger's home, which is located twelve blocks southwest of the park at 101 N 46th St. Apt 302. A variety of liquid refreshments will be provided. Potluck Director; Dave Enger

TUESDAY, NOVEMBER 25 - REDMOND INTRODUCTORY CLINIC An introduction to orienteering featuring a film, lecture and demonstration will be presented at Rose Hill Jr. High School in the Choral Room. The school is located at 13505 N.E. 75th St. in Redmond and the presentation is co-sponsored by the Redmond Parks and Recreation Department.

SATURDAY, DECEMBER 6 - FARREL-McWHIRTER PARK MEET A new area for COC located in a new city park located at 19200 NE 102nd Place in Redmond. The meet is co-sponsored by the Redmond Parks and Recreation Dept. Goran Larsson has designed the courses on a new 1:5,000 black and white map he has made.

MONDAY, DECEMBER 8 - CLUB MEETING & WORKSHOP Green Lake Library, 7364 E Green Lake Drive N in Seattle (north of Woodland Park). At 7:00 pm a workshop for beginners/novice orienters will be conducted and the club business meeting will begin at 7:30pm with a film on the 1976 Montreal Olympics.

SATURDAY, DECEMBER 27 - ST EDWARDS PARK MEET A great opportunity to work off some of that good holiday food and enjoy yourself doing it. The park is located near the northeast end of Lake Washington with the entrance on Juanita Drive at NE 145th St, south of Kenmore. Five color, 1:5,000 map.

SATURDAY, JANUARY 17 - WOODLAND PARK MEET The advance course will be score orienteering. In score orienteering each participant tries to visit as many controls as possible in any order within a given time limit. Each control has a point vlaue that is determined by it's distance from the start/finish as well as the degree of orienteering skill needed to locate it. Map; five-color 1:5,000.

SUNDAY, FEBRUARY 1 - LINCOLN PARK MEET Another new area for COC, located between Fauntleroy Way S.W. and Puget Sound in the southwest corner of Seattle. Meet near the main parking lot opposite SW Rose St. New 1:5,000 map and courses by Debbie Newell and Richard Lacy.

MONDAY, FEBRUARY 9 - CLUB ANNUAL MEETING Our most important meeting of the year, at which time we review the past year's accomplishments and elect officers for 1981. Mark your calendar for 7:00-9:00 at the Green Lake Library.

"Nuts! I probably just became the textbook example of poor route selection." (Cartoon by John Cassady)

MEET RESULTS

SEWARD PARK

OCTOBER 26, 1980

Mike Schuh and Debbie Newell directed a well-attended meet in Seward Park on October 26. Three courses were set, including a technically challenging orange that might have been deserving of being called a green course. Thanks go to Jean Davis, Dave Enger, the Larssons, and others for their help.

One minor problem developed when one competitor began to copy his course onto his map before being given a start time, much less having been officially started. This caused no real problem but everyone is reminded that besides being a bit unfair, it is adequate grounds for disqualification in sanctioned events.

RED COURSE 3.2KM 11 Controls

1.	Jeff Lewis, Oregon State Univ./Tulane	28:39
2.	Dave Enger, COC	36:10
3.	Goeran och Robert Larsson, COC	41:12
4.	Wayne Patterson, Oregon State Univ.	44:16
5.	Curt Ralston, Oregon State Univ.	44:54
6.	Bill Hall, COC	46:18
7.	Stephen Brigham, Oregon State Univ.	47:44
8.	Jean Davis, COC	50:03
9.	Berith Larsson & J.J., COC	52:31
10.	Dave Larson, COC	54:57
11.	Willard Sperry, EOC	78:55
12.	Lynne Pederson, COC	146:35

ORANGE COURSE 2.2KM 9 Controls

1.	Robert Miller, Seattle	39:28
2.	Harry Morgan, COC	44:35
3.	Gary & Debra MacDonald, COC	46:47
4.	Debbie Newell, COC	47:00
5.	Tom J. Lewis, Seattle	49:10
6.	Tom D. Lewis, Seattle	50:05
7.	Gabe Patterson, Seattle	50:10
8.	Harvey Friedman, COC	58:30
9.	Bob Barnes, COC	68:45
10.	James Bratt, EOC	77:30
11.	Jean Barker, Seattle	78:00
12.	Pat Malloy, EOC	80:20
13.	Barbara Ballard, EOC	87:15
14.	Debbie Bostrum, EOC	91:00
15.	Anne York, COC	92:35
16.	Joanne Sperry, EOC	104:35
17.	Jim Moy, COC	109:43
18.	Sandy Kish-Miller, COC	114:35
19.	Cecilia Werner, EOC	123:00
20.	Mabel Moy, COC	124:03
21.	Bill & Gene Fritz, COC	146:35
22.	Larry Dunlap, Seattle	147:00
	Martha Brinkerhoff, COC	DNF

WHITE COURSE 1.2KM 5 Controls

1.	Robert Miller, Seattle	12:02
2.	John Lilly, Seattle	13:30
3.	Richard Farmer, Bellevue	17:10
4.	Sandy Kish-Miller, COC	18:18
5.	Nancy Lewis, Seattle	20:15
6.	Debra Lewis, Seattle	22:15
7.	Pam & Joann Stevens	27:20
8.	William Turner, COC	28:20
9.	Bob & Jenny Koening, Kirkland	30:25
10.	Janet Danielson, Seattle	34:25
11.	Dave Meyer, Edmonds	36:00
12.	Mabel & Gordon Pederson, Seattle	36:15
13.	Marion Koening, Kirkland	39:28
14.	Ellen Roth	43:39
15.	Tom Moy, COC	44:30
16.	Jean Barker, Seattle	44:50
17.	Tim Simons, Auburn	46:10
18.	Jo Ellyn Kuipers, COC	46:30

CLUBS —

COC Cascade O Club
EOC Ellensburg O Club
NSO North Shore Orienteers (BC)
OUBC Orienteering UBC

STEVENS PASS

Gary MacDonald and Mike Schuh presented a brief workshop and set a short course for a Pacific Northwest Professional Ski Patrol Association training weekend at Stevens Pass on a sunny October 19. Though poorly attended because of its place on the agenda (last), the event was enjoyed by all who participated. The Stevens Pass ski area is a possible site for future club meets in late summer or early autumn.

Course: Yellow Open
8 controls
1.4 km

1.	Ed	19:15
2.	Mike	22:35
3.	Sally & Doris	39:00

CARKEEK PARK

NOVEMBER 8, 1980

RED COURSE 4.5KM 14 Controls

1.	Richard Lacy, COC	61:51
2.	Hendrick Nordholm, COC	62:37
3.	Egon Nordholm, COC	64:19
4.	Robert Miller, Seattle	73:57
5.	Jeff Kelley-Clarke, COC	85:03
6.	David G. Larson, COC	87:57
7.	Eric D'Asaro, COC	93:16
8.	Gail Hanna, SDOC	115:58
9.	William Connett, COC	156:40

ORANGE COURSE 2.8KM 9 Controls

1.	Paul Gauthier, Bellevue	54:30
2.	Jim Finn, EOC	60:52
3.	Debbie Newell, COC	61:55
4.	Harvey Friedman, COC	69:54
5.	Jean Barker, Seattle	78:14
6.	Fred Littel +1, Seattle	86:17
7.	Jim Moy, COC	94:03
8.	Judith Connett, COC	97:19
9.	Mabel Moy, COC	117:01
10.	Jeanitt Nordholm, COC	117:30
11.	Patricia Malloy, COC	125:33
12.	Lynne Pedersen, COC	133:30
13.	Debra & Lorna MacDonald, COC	135:03
14.	Janet Danielson & Renee' Olbricht	141:55

YELLOW COURSE 1.9KM 8 Controls

1.	Carol Johnson, COC	48:29
2.	Alan Stay, Seattle	65:54
3.	Anne York, COC	70:01
4.	Sandra Fish-Miller, COC	74:16
5.	Tom Moy, COC	75:25
6.	Nancy Lewis, Seattle	103:47
	Dave Meyer, Edmonds	DNF

WHITE COURSE 1.1KM 6 Controls

1.	Al Riggs, Edmonds	39:48
2.	Dave Meyer, Edmonds	41:13
3.	Alan Stay, Seattle	43:31
4.	Kurt Oberloh & group, Seattle	55:15
5.	Marie Larson + 3, Sumner	59:00

Some Thoughts on Making and Reading Maps

Besides being a great favor to other orienteers in your area, making an O map can really improve your orienteering. If you've made maps, you know what I mean; for those who haven't, let me try to explain.

All maps are fallible - they all have errors and omissions. Some are better than others, some a lot better, but all have mistakes. The mistakes occur for two reasons even on good maps: 1) The they're made by humans and we have a knack for doing imperfect jobs. In the process of field checking, redrawing, and printing it's remarkably easy to leave something off, or say, interpret a speck of dirt as a boulder. 2) Map making is very subjective, for example; how do you map a marsh that is flooded (and looks very "uncrossable") one day during a rainstorm, but is almost dry when you go by it a day later?

So the map you end up using represents one person's interpretation (or several persons different interpretations) of the terrain, for those parts of the map they actually field-checked and accurately drafted. And then there are those areas or whole maps that haven't been field checked! This may sound pretty discouraging, but it shouldn't. If you are aware of the possible problems a map, any map, can have, then you can deal with them. There are several ways to do this.

First, when competing on a new map, start judging its accuracy as soon as you can - before your start time if maps are given out early. Examine the terrain and the map from a map maker's point of view. Are the trails near the start/finish correct, what size boulders (if any) are mapped, do the contours look right or are they very generalized, and so on? Sometimes part may be good and part bad - the trails may be just right, but the contours terrible. Figure out how much you can rely on the map, or how little. Since it normally takes a leg or two on a course to do this, that's a good reason to take the first couple of legs carefully. (Beginning to see where some map making experience can improve your orienteering?)

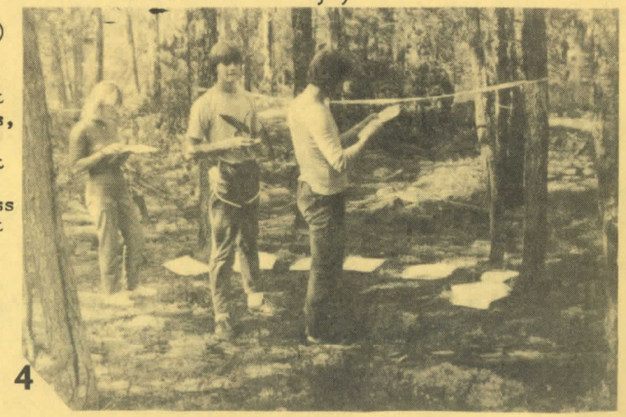
Second, never rely on just one feature on the map for your navigation - even on an excellent map it may be the one item in a thousand that got misdrawn. (Likewise, you never use just a compass, or just pace-counting) Always have two or three safety checks to make sure you're going the right way - other features on the map that you will notice as you go by them, as well as rough compass and pace checks. It doesn't take much more effort or time and if you do go the wrong way you will realize it a lot sooner and can correct yourself with little loss of time.

Third, if it's a lousy map, you can still ori orienteer. It's just more important to use your compass and your pace-count and still get what you can out of the map. For example, if the trails are not accurate, don't just look for, say the second trail on the left. Look instead for, say the second trail on the left in about 400 meters that goes sharply uphill to the northeast. If the second left comes in 150 meters and goes downhill to the southeast then you won't just follow it blindly. Rather you'll stay on course and make a mental note of anoth trail that wasn't on the map. Remember - good map or bad, it's the same for everyone.

Finally, if you've made a map and realize how much time and effort goes into it, you'll be a lot less eager to criticize anyone else's efforts.

— Peter Gagarin, Control Point 6/7-77

Egon Nordholm, Glen Englund and Jeanitt Nordholm all taking off at a meet in B.C. (above) while at the map location (below) Glen was not too concerned about studying his map.



SYMBOLS FOR A BLACK AND WHITE ORIENTEERING MAP

Robin Harvey in his book Map Making for Orienteers designed this list of symbols for new mappers to use when they want to try a simple black and white map and at the same time attempt to achieve the quality found in color map symbols. With the absence of color the difference between symbols must be shown with line thickness and the pattern used. For the time being we will overlook line thickness because most beginning mappers do not have technical pens available to them or wish to purchase them. (Mappers who would like to maintain Harvey's recommended standard in line thickness can talk with me for more information) A quick look at the list will show that there are fewer symbols than what is used on a color map. This is necessary to avoid confusion between similar color symbols and at the same time keep the map easy to read in the absence of color. For example, a black dotted line is used exclusively for water features (streams and ditches) on a black and white map to avoid confusion with contour lines or a road. A mapper must keep in mind that an orienteer's route selection is based on the information provided on the map. It must not only be accurate but legible as well.

1.4	major road (optional)
0.7	road
0.7	cart track
0.35	footpath
0.18	narrow ride
0.18	wide ride
0.18(0.25)	uncrossable fence
0.18(0.25)	crossable fence
0.18(0.25)	wall
0.18(0.25)	ruined wall earth bank
0.7	railway
0.25	stream
0.18	ditch
LF 151 M	lake
LT 111	uncrossable marsh
LT 70	crossable marsh

0.18	contour
0.18	form line
S	knoll
S	pit
0.18	gully
0.18	steep slope
0.7	dangerous cliff
S	outcrop
S	boulder
LT 23	open land
LT 97	light thicket
LT 99	dense thicket
S	platform
S	building

Symbols to be found on the sheet of self adhesive map symbols, available from the BOF National Office, are marked with the letter 'S'.

FT LEWIS CLASS B MEET

BLUE COURSE 10.2KM 16 Controls

1.	Egon Nordholm		88:12
2.	Reidar Zapf-Gilje	OUBC	95:55
3.	Bruce Rennie	NSO	100:37
4.	Ole Anker-Rasch	NSO	101:49
5.	Mike Ackerfeldt	NSO	105:23
6.	Dave Enger		107:11
7.	D.G. Larson		148:00
8.	David E. Larson		148:08
9.	Bill Hall		150:28
	Afan Jones		(81:19) DNF #13
	Richard Lacy		DNF
	Dave Rudakewich		DNF
	Storakers		DNF

RED COURSE 6.1KM 12 Controls

WOMEN'S CLASS

1.	Robyn Rennie	NSO	61:31
2.	Jean Davis		83:40
3.	Berith Larsson		88:20
	Lynne Redersen		DNF

MEN'S CLASS

1.	Pat Berting		67:41
2.	Hemrik Nordholm		72:56
3.	Charlie Bock		73:03
4.	Glen Englund		155:19
5.	Willard Sperry	EOC	170:37
6.	Tom Hardin		179:32
	Tom Louis		DNF
	Belanger		DNF

ORANGE CRS 3.9 KM 7 Controls

WOMEN'S CLASS

1.	Debbie Newell	67:32
2.	Sue Hamilton	118:13
3.	Cecilia Werner	118:48
4.	Diane Krause	140:50
5.	Mabel Moy	219:17
	Jeanitt Nordholm	DNF
	Martha Brinkerhoff	DNF

MEN'S CLASS

1.	M/M Mac Donald	55:46
2.	Jack Keegan	61;46
3.	Ernie Priestley	61:49
4.	A. Farash	76:04
5.	Jerry Graham	111:45
6.	M/M Powell	115:17
7.	Bill Maiden	116:30
8.	Harvey Friedman	119:40
9.	Michael Mayo	156:44
10.	M/M Hendrickson	167:11
	Len England	DNF

WHITE COURSE 2.9KM 6 Controls

WOMEN'S CLASS

1.	Diane Kraus	54:08
2.	Lavon Kraus	72:59
3.	Breean Beggs	
3.	Lois Larson	80:49
4.	Jean Barker	93:08

MEN'S CLASS

1.	Gabe Patterson	34:13
2.	Jim Moy	34:19
3.	Hudson & Miller	38:24
4.	Fritz Family	68:40
5.	M/M Dederson	70:26
6.	Stan & Karen	78:54
7.	Troop 636	99:25